CODEAXIS HACKATHON ROUND 3: THE FINAL BUILD

BRANDTRACE

Evidence-Aware AI-Assisted Brand Governance Platform

Team: Team_Nexus | Repository: C:\Projects\BRANDTRACE\BRANDTRACE | Date: 2026-10-30

CORE PRODUCT PRINCIPLE:

"Every finding has a source. Every decision has an owner."

BRANDTRACE insists that compliance findings link directly to active approved brand rules and facts. High-risk violations programmatically block publication sign-off at the server level until an authorized human records a documented disposition.

VERIFICATION VECTOR	GROUND-TRUTH RESULT	STATUS & EVIDENCE
Backend Test Suite	**133 passed, 1 skipped** (0 failed across 18 test files)	126 baseline + 7 Round 3 acceptance suites; pytest Python 3.10
Automated Browser QA	**32 / 32 PASSED (100%)**	Playwright Chrome 156 against dedicated isolated backend 8031 & frontend 3001
Isolated PostgreSQL 18 Live	**ALL 11 PHASES & 4 PYTESTS PASSED (100%)**	36 assertions verified on disposable cluster 5438; host 5432 & dev DB untouched
Hosted Neon Staging Runbook	**10 / 10 CHECKS PASSED (100%)**	Neon PostgreSQL port 8017 verified; rule_6db288b87ba4 retained artifact
TypeScript & Production Build	**0 errors · 51,297 chars prerendered**	1,691 modules transformed; production bundle verified via npm run build
API Route Inventory	**40 registered routes (100% accounted)**	31 user-facing UI active, 5 internal admin, 2 demo lifecycle, 2 diagnostics

1. Product Workflow & Server-Enforced Governance Gate

BRANDTRACE reviews content drafts (plain text, markdown, PDF documents, or raster images) against deterministic brand rules and time-aware facts. The application enforces a rigorous five-stage state machine that prevents unverified content publication:

Workflow Stage	Engine Action & Architectural Enforcement	Resulting Record State
1. Submission & Ingestion	Draft submitted via Direct Text, PDF upload (pypdf), or Image upload. 64KB bounded streaming size guard prevents buffer overflow attacks (HTTP 413 guard).	Status: IN_REVIEW Initial version snapshot created in review_versions.
2. Deterministic Analysis	Evaluates prohibited phrases, required product terms, fact validity dates, and CIELAB color distances. Links every finding to a source rule/fact.	Clearance Score computed (0–100). Status: CONTRADICTED, OUTDATED, NEEDS_REVIEW.
3. Evidence Decision Chain	Reviewer inspects flagged passage, triggered rule, and grounded fact. Can Accept (exact-span replacement, 0 word duplication), Edit, or Dismiss.	Finding status: RESOLVED. Dismissal strictly requires human reason text.

Workflow Stage	Engine Action & Architectural Enforcement	Resulting Record State
4. Approval Gate Boundary	Server-enforced boundary: POST /api/reviews/{review_id}/approval checks for unresolved HIGH/CRITICAL findings. Returns HTTP 409 Conflict if blocked.	Draft remains IN_REVIEW if blocked. Transitions to APPROVED once cleared.
5. Audit & Rollback Ledger	Every mutation appends an immutable compliance record to audit. Prior revisions can be restored with automatic compliance re-analysis.	Immutable version history maintained; zero destructive in-place overwrites.

1.1 Human-in-the-Loop Governance State Transitions & Approval Guardrails

To prevent unverified claims or silent bypass of corporate policy, BRANDTRACE enforces mathematical clearance scoring and strict transactional gate conditions across every draft review lifecycle:

Governance Vector	Mathematical Specification & Rule Logic	Operational Verification Guard
Clearance Score Formula	Score = max(0, 100 - Sum(Penalties)) • CRITICAL: -25 pts | HIGH: -15 pts • MEDIUM: -8 pts | LOW: -3 pts • Monotonically decreasing penalty model ensures any unaddressed violation visibly drops draft health.	Deterministically evaluated on review ingestion, finding resolution, and version restore.
HTTP 409 Approval Gate	Draft status transition from IN_REVIEW to APPROVED is strictly gated at the server API layer: if unresolved_high_or_critical > 0: raise HTTPException(409, 'Approval blocked'). UI approval button is disabled, and direct API calls receive HTTP 409 Conflict.	Enforced in backend/app/main.py (approval endpoint). Verified in automated regression suites.
Finding Resolution Lifecycle	1. **NEEDS_REVIEW**: Default state on ingest. 2. **ACCEPTED**: Reviewer accepts suggested correction (exact-span string replacement, zero duplication). 3. **DISMISSED**: Reviewer overrides finding (strictly requires documented reason text in audit). 4. **ESCALATED**: Routed to Brand Administrator for constitutional guideline amendment.	Every state change appends a timestamped, actor-attributed event to the append-only audit ledger.

2. System Architecture & Relational Persistence

BRANDTRACE operates a decoupled, enterprise architecture connecting an Apple-inspired React 19 frontend to an asynchronous FastAPI backend. Persistence is abstracted via SQLAlchemy, supporting seamless portability between local SQLite development and PostgreSQL 18 / hosted Neon production.

Layer / Component	Technologies & Libraries	Key Responsibilities & Operational Guarantees
Frontend Client	React 19, TypeScript, Vite 6.4.5, TailwindCSS, Lucide-React	Single-page workspace with 9 specialized views, interactive SVG analytics, responsive navigation (375px mobile to 4K desktop), and accessible contrast.
API Client Layer	Typed REST client (src/services/api.ts)	Provides 34 typed client SDK methods spanning all user-facing, demo, and diagnostic routes with role simulation headers; 5 admin routes managed via CLI/runbooks.
API Service Boundary	FastAPI 0.115, Uvicorn, Pydantic v2	Strict request validation, Bearer/API-key token auth in production mode, tenant isolation, and centralized HTTP error mapping (401, 403, 409, 413, 422).
Governance Core	Python 3.10 standard library, pypdf, Pillow	Deterministic exact-span regex matching, CIELAB Delta-E color distance, W3C WCAG 2.1 relative luminance, and cryptographic SHA-256 byte hashing.
Relational Persistence	SQLAlchemy 2.0, Psycopg 3, SQLite & PostgreSQL 18	15 normalized tables: workspaces, workspace_members, system_state, rules, rule_versions, facts, fact_versions, brand_profiles, brand_assets, reviews, review_versions, findings, evidence_dependencies, rule_suggestions, audit.

Relational Schema Integrity & Workspace Tenant Isolation

All data-bearing tables enforce workspace isolation via workspace_id columns. In Phase 1 audit remediation, guideline-health and rule-suggestions were updated to scope findings through their parent reviews (LEFT JOIN reviews r ON f.review_id = r.id). Foreign-workspace queries and writes are rejected with HTTP 403 Forbidden with zero database modifications. Demo reset (POST /api/demo/reset) is defensively constrained to workspace_id = 'ws_default', safeguarding other workspace partitions.

3. Three Major Round 3 Commitments: Verified Implementations

Commitment	Architectural Implementation & User-Facing Path	Acceptance Test & Verified Proof	Limitations Disclosed
1. Time-Aware Claim Ledger & Impact Map	Facts carry validity windows (effective_from, valid_until) and notes. 30-day expiring-soon badge and hard expiration alerting. Bi-directional dependency tracking in evidence_dependencies. Interactive SVG impact graph. Targeted re-review (POST /api/governance/re-review-impact).	**test_scenario_a_claim_ledger_acceptance.py (PASS)** Verified: 1 approved fact expired with 3 dependent drafts + 1 unrelated draft. Exactly the 3 dependent drafts transitioned to CHANGES_REQUESTED; unrelated draft stayed APPROVED.	Impact graph relies on explicitly tracked rule/fact links; arbitrary unlinked semantic inference is not attempted.
2. Multimodal Visual Brand Compliance	PDF text extraction via pypdf. 64KB bounded streaming upload guard. Pillow-based dominant palette extraction with CIELAB Delta-E color distance against visual brand profile. W3C WCAG 2.1 relative luminance contrast on design tokens. Cryptographic SHA-256 asset registry (GET/POST /api/brand-assets, DELETE /api/brand-assets/{asset_id}).	**test_scenario_b_visual_compliance_acceptance.py (PASS)** Verified: 6 design-token pairs evaluated with 100% precision & recall (TP=3, TN=3, FP=0, FN=0). Asset registry flags exact binary matches (APPROVED/RESTRICTED).	Upload-derived contrast is reference-only (NEEDS_REVIEW) due to indeterminate text roles. Pure-raster OCR is unsupported (Option B).
3. Guideline-Health Intelligence & Feedback	Aggregates finding telemetry, dismissals, and reviewer disagreements. Distinct-actor disagreement detection. Traceable clarification proposals (GET /api/governance/rule-suggestions). Reviewer triage (POST /api/governance/rule-suggestions/{rule_id}). Admin activation (POST /api/governance/rule-suggestions/{suggestion_id}/activate) with optimistic concurrency (HTTP 409).	**test_scenario_c_guideline_health_acceptance.py (PASS)** Verified: Disagreement detected across 3 independent drafts. Reviewer proposal acceptance rejected (HTTP 403); admin accepted (HTTP 200). Stale version rejected (409); valid activation promoted rule to v2 with audit.	Suggestions provide deterministic advisory text; automated policy mutation without administrator sign-off is prohibited.

4. Comprehensive Verification Evidence & QA Proof

Every layer of BRANDTRACE has been subjected to rigorous, repeatable automated testing using dedicated isolated processes and disposable databases. The live development database (brandtrace.db) and host services remained 100% untouched throughout all runs.

Test Suite / Harness	Target & Execution Mode	Coverage & Assertions Verified	Result & Proof
Standard SQLite Regression (scripts/verify_sqlite_regression.py)	Dedicated disposable SQLite database; 14 test modules	Baseline API endpoints, color science, PDF parser, production auth, disposition integrity, QA regression, claim ledger, visual benchmark, guideline health, defect repairs.	**126 passed, 1 skipped** (22.8s runtime, 0 failed)
Round 3 Security & Acceptance (backend/tests/test_scenario_*.py)	Ephemeral pytest process; 4 test suites	Phase 1 security access control (401 unauth, 403 foreign workspace, 0 writes), Scenario A (claim ledger & impact re-review), Scenario B (visual compliance & contrast), Scenario C (guideline health & activation).	**7 passed, 0 failed** (2.36s runtime)
Combined Backend Pytest	All 18 test files in backend/tests	Full test suite across both baseline and Round 3 security/scenario acceptance suites.	**133 passed, 1 skipped** (1 skipped requires Gemini key)
Isolated Browser QA Suite (scripts/run_isolated_browser_qa.py)	Headless Chrome 156; isolated backend port 8031, Vite frontend port 3001, disposable DB	Full 32-scenario end-to-end journey: Hero navigation, tour, draft submission, neutral corrections, missing terms, broad claims, contamination guards, approval gate (409), dismissal reasons, PDF upload, image palette, role simulation, asset registry, impact map, guideline health, fact notes, profile editing, rule version history, empty/error states, cross-workspace denial (403).	**32 / 32 PASSED (100%)** (56.4s runtime, 0 failed)
Isolated PostgreSQL 18 Live (scripts/test_postgresql_isolated_live.py)	Disposable PostgreSQL 18 cluster (port 5438); SCRAM-SHA-256 auth; strict ASSERT_POSTGRESQL=1	15 public tables schema initialization, idempotency, bootstrap lifecycle (401, 403, 201, 409), core CRUD & JSON datatypes, strict cross-workspace isolation (ws_alpha vs ws_beta zero leakage), review lifecycle & approval gate, transaction rollback, and 4 pytest suites (36 live PG assertions).	**ALL 11 PHASES PASSED** (Host port 5432 & dev DB untouched)
Hosted Neon Staging Runbook (scripts/verify_hosted_staging.ps1)	Hosted Neon PostgreSQL cluster; dedicated port 8017	10 acceptance checks: schema verified, /api/health and /api/ready HTTP 200, unauth 401, reviewer bootstrap 403, idempotent bootstrap 409, rule creation 201, rule version history 200, foreign workspace 403, member provisioning 201, lockout prevention 422. Retained acceptance artifact rule_6db288b87ba4.	**10 / 10 CHECKS PASSED** (Clean run, process stopped)
Frontend Static Typecheck (npx tsc --noEmit)	TypeScript compiler (Node v20)	Strict type validation across all React components, analytics charts, modals, and API client interfaces.	**0 errors**
Production Build & Prerender (npm run build)	Vite 6.4.5 + TSX prerenderer	Production bundle compilation, chunk optimization, gzip size computation, and static HTML prerendering.	**Passed** (1,691 modules transformed, 51,297 chars prerendered)

5. Apple-Inspired UI Design & Analytics Architecture

BRANDTRACE features a cohesive, light enterprise design system inspired by the precision, whitespace, and typographic restraint of Apple, Mitigata, and Gordon. The interface balances clean executive presentation with deep evidentiary inspectability:

Component / View	Design Execution & Interactive Capability	Evidentiary Truth & Data Binding
BrandtraceHero & Evidence Prism	Interactive marketing landing page with live dynamic /api/ready status probe, 15-second value proposition, and two primary CTAs: 'Enter Workspace' and 'Take Judge Tour'.	Dynamic health badge; displays live database dialect from server. Zero hardcoded passing test claims.
Workflow Stepper (WorkflowStepper.tsx)	5-stage interactive progression stepper linking Brand Constitution, Content Review, Evidence Triage, Governance Approval, and Guideline Telemetry.	Reflects real workflow stages; allows one-click navigation across workspace views.
Review Outcome Donut (ReviewOutcomeDonut.tsx)	Interactive SVG donut chart displaying distribution across Approved, In Review, Changes Requested, and Waived drafts with hover tooltips and clearance pass rates.	Directly computed from reviews array; reflects real database status counts.
Severity & Status Bar Charts (SeverityBarChart.tsx, EvidenceStatusBarChart.tsx)	Proportional stacked bar charts showing findings by severity (CRITICAL, HIGH, MEDIUM, LOW) and evidence status (CONTRADICTED, OUTDATED, UNSUPPORTED, NEEDS_REVIEW).	Directly bound to insights.by_severity and insights.by_evidence_status from /api/insights.
Activity Timeline (ActivityTimeline.tsx)	Chronological provenance timeline showing recent submissions, approvals, waivers, and rule updates with actor badges and relative timestamps.	Bound to real audit records and reviews metadata; no synthetic events.
Interactive Impact Graph (InteractiveImpactGraph.tsx)	SVG dependency graph linking verified facts to dependent drafts. Displays expiring-soon alerts, hard expiration badges, and includes a tabular list fallback.	Constructed strictly from evidence_dependencies, facts, and reviews.
Evidence Decision Chain (EvidenceDecisionChain.tsx)	Four-node provenance progression rendered on every finding card: 1. Flagged Passage → 2. Policy Rule → 3. Grounded Evidence → 4. Human Disposition.	Displays actual passage text, rule ID, fact excerpt, source URL, and recorded reviewer reason.
Guided Judge Tour Modal (JudgeTourModal.tsx)	Step-by-step interactive walkthrough guiding evaluators through the 7 core pillars of BRANDTRACE with keyboard navigation (Esc, arrows) and view navigation.	Informational walkthrough; performs zero destructive mutations during tour navigation.

6. Security Architecture & Honest Technical Limitations

Security, RBAC, and Authentication Boundaries

• **Production vs Demo Mode:** When ALLOW_DEMO_AUTH=False, all synthetic demo headers (X-Demo-Role, X-Demo-Actor) are strictly rejected with HTTP 401. Production requests require Bearer tokens or verified API keys.
• **Permanent One-Way Bootstrap:** POST /api/bootstrap records a durable completion marker in system_state. Subsequent bootstrap attempts return HTTP 409 Conflict.
• **Lockout Prevention:** Workspace administrators cannot remove their own membership (HTTP 422 Unprocessable Entity), preventing accidental organizational lockout.
• **Zero Client Secrets:** Privileged database credentials, admin API keys, and server-side connection strings are never exposed in frontend bundles or VITE_* environment variables.

Transparent Disclosure of Technical Boundaries

Capability Area	What BRANDTRACE Implements	Explicit Limitation / What Is NOT Claimed
Visual Contrast Analysis	W3C WCAG 2.1 relative luminance calculation on semantic foreground/background design token pairs (verified across 6 benchmark pairs with 100% precision/recall).	Dominant-color ratios extracted from uploaded raster images are heuristic indicators labelled NEEDS_REVIEW. Automatic WCAG text pass/fail is NOT claimed for unsegmented graphics.
Brand Asset Verification	Cryptographic SHA-256 binary hash matching against registered brand assets with status tracking (APPROVED, RESTRICTED, NEEDS_REVIEW).	Does NOT perform semantic vector matching, scale/rotation-invariant computer vision, or perceptual logo identification.
Document & Image OCR	Pure-Python text layer extraction from PDF documents via pypdf with comprehensive corrupt/blank/encrypted file handling.	Pure-raster OCR is intentionally UNSUPPORTED (Option B). The system emits explicit ocr_unsupported notices rather than fabricating heuristic extractions.
Multi-Tenant Architecture	Single-enterprise multi-workspace partitioning (workspace_id scoping) with server-enforced tenant boundaries.	Does NOT provide public multi-tenant SaaS isolation with self-service tenant registration. Full multi-tenant SaaS requires external OIDC/OAuth2 IdP federation.

7. Team Contributions & Submission Package

Metadata Attribute	Submission Detail & Verification Reference
Hackathon / Round	**CodeAxis Hackathon — Round 3: The Final Build**
Project Name	**BRANDTRACE — Evidence-Aware AI-Assisted Brand Governance Platform**
Team Name	**Team_Nexus**
Core Deliverables	1. Time-Aware Claim Ledger & Impact Map | 2. Multimodal Visual Compliance | 3. Guideline-Health Intelligence
Backend Test Verification	**133 passed, 1 skipped** across 18 test files (pytest, Python 3.10.11)
Browser QA Verification	**32 / 32 PASSED (100%)** (Playwright Chrome 156 against isolated ports 8031/3001)
PostgreSQL Verification	**ALL 11 PHASES PASSED** (PostgreSQL 18 live cluster) | **10/10 CHECKS PASSED** (Hosted Neon staging)
Production Build Verification	**0 errors** (TypeScript tsc --noEmit) | **51,297 chars prerendered** (Vite production build)
Working Tree State	Clean working tree verified on commit 3d2539c. No uncommitted modifications or secrets.
Remaining Manual Actions	User deployment to public hosting (Vercel/Render), upload of this PDF to Google Drive, and submission of the official Round 3 Google Form.